BASE METRICS

ENTERPRISE STORAGE ARCHITECTURE & SCALING BLUEPRINT

Comprehensive Multi-Sector Technical Guide: 5,000+ Users, Qualitative Audio, Photos & GPS

1. EXECUTIVE SUMMARY & DUAL-TIER STORAGE ARCHITECTURE

Base Metrics is engineered to handle rigorous, high-throughput field data collection across diverse sectors, including Education, Healthcare, Agriculture, WASH, and Humanitarian aid. To support 5,000+ active field users without performance bottlenecks or excessive costs, Base Metrics employs an industry-standard Dual-Tier Cloud Architecture:

- **Tier 1: Cloud Firestore (NoSQL Operational Database):** Houses all lightweight structured documents, questionnaire responses, student attendance, enumerator rosters, audit logs, and locked GPS coordinates. Firestore scales effortlessly to millions of documents with sub-second query latency.
- **Tier 2: Google Cloud Storage (Media & Qualitative Object Storage):** Houses all binary media files, including high-resolution photographic evidence, voice recordings, and qualitative audio interviews. Only the lightweight HTTPS URL string (100 bytes) is written to Firestore.

2. THE 1 MB FIRESTORE DOCUMENT RULE & DATA SIZING

A fundamental rule of Google Cloud Firestore is that every single document has a hard maximum size limit of 1 MB (1,048,576 bytes). Understanding this boundary is key to maintaining a high-performance system:

Data Category	Average Payload Size	Documents to Reach 1 MB	Firestore Capacity
Structured Survey / Attendance	1 KB - 4 KB	250 - 1,000 surveys	Virtually Unlimited
High-Precision GPS Lock	80 - 120 Bytes	10,000+ GPS points	Negligible footprint
Compressed Verification Photo	60 KB - 120 KB	8 - 14 photos	Exceeds limit if stacked
1-Minute Audio Voice Note	500 KB - 1.2 MB	1 - 2 files	Breaches 1 MB limit
15-Minute Qualitative Interview	5 MB - 15 MB	Direct breach	MUST use Cloud Storage

Key Architecture Rule: By decoupling binary files (photos and audio) from Firestore documents and hosting them in Cloud Storage, each Firestore survey document stays under 5 KB. This guarantees that 1-click CSV exports remain ultra-fast and Excel/SPSS never choke on oversized payloads.

3. 5,000 USER CAPACITY CALCULATIONS & COST PROJECTIONS

The following table outlines monthly storage, bandwidth, and cost estimates for a deployment of 5,000 active field workers submitting an average of 5 reports per day (25,000 records daily / 750,000 monthly):

Deployment Tier	Monthly Records	Firestore DB Size	Cloud Media Size	Estimated Cost
Structured Text + GPS Only	750,000 reports	2.25 GB / month	0 GB	$0.40 - $2.50 / mo
Text + GPS + 1 Photo/Audit	750,000 reports	2.25 GB / month	60.0 GB / month	$2.00 - $4.00 / mo
Text + GPS + Photo + Audio	750,000 reports	2.25 GB / month	250.0 GB / month	$7.50 - $12.00 / mo

- **Firestore Storage Cost:** Google Cloud charges $0.18 per GB per month for Firestore database storage.
- **Cloud Storage Cost:** Google Cloud Storage standard tier costs only $0.026 per GB per month (over 85% cheaper than database storage). Storing 100 GB of field photos costs less than $2.60 per month.
- **Network Bandwidth:** Client-side compression ensures field personnel consume minimal mobile data bundles (less than 150 KB per full report with photo and GPS).

4. QUALITATIVE DATA COLLECTION & AUDIO INTERVIEWS

Qualitative data collection often requires rich, verbatim open-ended responses from community elders, clinic head nurses, or beneficiary focus groups. Base Metrics is prepared to support audio data collection through the following pipeline:

- **Browser MediaRecorder API:** Captures crystal-clear voice notes directly from mobile browsers or PWA without requiring external third-party apps.
- **Modern Codec Compression:** Audio is encoded using Opus / AAC (16 kHz, mono), producing high-clarity voice recordings at a compact ~8 KB/sec (only ~500 KB per minute).
- **Automatic Speech-to-Text Transcription:**Audio recordings can be fed directly to the Gemini API or Whisper transcription backend. The resulting text transcript is stored in Firestore, enabling instant keyword search, sentiment tagging, and full thematic qualitative analysis.
- **Qualitative Text Fields:**For typed qualitative responses (e.g. detailed case notes, beneficiary testimonials), Firestore string fields support up to 500,000 characters per field without any external storage needed.

5. PHOTOGRAPHIC AUDIT & GPS TAMPER-PROOFING

For field verification, physical evidence is vital to ensure enumerators are legitimately on-site:

- **Client-Side Image Canvas Compression:** PhotoCapture automatically resizes mobile camera uploads to a maximum 960px dimension at 72% JPEG quality. This reduces an 8 MB raw camera snapshot down to 70 KB in milliseconds before transmission.
- **Programmatic GPS Locking:** High-accuracy GPS coordinates (latitude, longitude, horizontal accuracy radius in meters) are locked programmatically via the HTML5 Geolocation API at the moment of submission.
- **Anti-Spoofing Protocol:** GPS timestamps are cross-referenced with server-generated Firestore request timestamps, preventing device clock tampering.

6. OFFLINE OUTBOX & DEVICE LOCAL STORAGE (RURAL OPERATION)

In remote communities where cellular connectivity is nonexistent, Base Metrics provides complete offline operational reliability:

- **Offline Storage Hierarchy:** Modern mobile browsers provide 5 MB - 10 MB in localStorage and 50 MB - 2 GB in IndexedDB (up to 20% of device storage).
- **Offline Capacity Per Phone:** A single facilitator smartphone can store over 10,000 structured text reports, or 50 to 100 complete reports with compressed photos, completely offline.
- **Automatic Queue Flushing (Outbox Sync):** The built-in service worker and offline sync manager monitor network events. As soon as the enumerator returns to a network zone, the app automatically flushes the outbox queue in batches without any manual intervention.

7. ENTERPRISE DATA GOVERNANCE & SECURITY

All multi-sector data collected through Base Metrics is protected by enterprise-grade controls:

- **End-to-End Encryption:** Enforced TLS 1.3 encryption in transit and AES-256 cloud encryption at rest on Google Cloud.
- **Role-Based Access Control (RBAC):** Strict server-side Firestore security rules partition data between Facilitators, Supervisors, B1 Collectors, and Administrators.
- **One-Time Access Codes & Master Gate:** Every field user requires an approved one-time activation code. Admin command center features are secured behind master gate verification.
- **Google API Limited Use & GDPR Compliance:** Strict compliance with Google API Services User Data Policy: zero advertising, zero third-party data broker sharing, and guaranteed data deletion requests within 48 business hours.